Avi Vajpeyi

avivajpeyi.github.io
github.com/avivajpeyi

Education

Ph.D. in Astrophysics (Advisors: Dr. R. Smith, Prof. E. Thrane) — Monash University
Thesis: "Astrophysical Inference in the Data Driven Era" — 2019–2022

B.A. in Computer Science and Physics (GPA: 4.0/4.0, Valedictorian) — The College of Wooster
Thesis: "Accelerating Granular Flow on GPUs with OpenCL & CUDA" — 2014–2018

Selected Research

Multivariate P-spline spectral estimation — arXiv preprint (2026)
Flexible spectral-matrix estimation for correlated multivariate time series. — arXiv:2607.04833

Differentiable WDM transform — arXiv preprint (2026)
Differentiable time-frequency methods for gravitational-wave and LISA analysis. — arXiv:2606.20269

Variational inference for correlated noise — Physical Review D (2025)
Scalable Bayesian inference for correlated gravitational-wave detector networks. — PRD 111, 062003

Bayesian P-spline modelling of LISA noise — Physical Review D (2026)
Non-parametric spectral-density estimation using penalised splines. — arXiv:2510.00533

Evidence estimation via density approximation — Physical Review D (2026)
Evidence estimation using informed approximations to posterior densities. — arXiv:2512.10283

For a complete publication record, see Google Scholar.

Industry Experience

Software Consultant — Quaisr
Built data-agnostic ML/active-learning blocks linking TensorFlow backends to visual workflows. — 2022 - 2023

Software Engineer — Bamboo Learning Inc
Developed NLP components and maths/listening-comprehension content for Alexa. — July 2018 - March 2019

Machine Learning Consultant — MOAD
Deployed CNN image classifiers and delivered hands-on machine-learning tutorials. — July - Aug 2018

Software Engineering Practicum — Gitkeeper
Built and tested Python backend components for Git-based automated assessment. — Aug 2016 - Feb 2017

Publications

[1] N. Aimen, P. Maturana-Russel, **A. Vajpeyi**, N. Christensen, and R. Meyer, "Bayesian power spectral density estimation for LISA noise based on penalized splines with a parametric boost", *Phys. Rev. D*, Jan. 2026. arXiv: 2510.00533 `[gr-qc]`.

[2] A. Boumerdassi, M. C. Edwards, **A. Vajpeyi**, and O. Burke, "First-time assessment of glitch-induced bias and uncertainty in inference of extreme mass ratio inspirals", *Phys. Rev. D*, 2026. arXiv: 2512.16322 `[gr-qc]`.

[3] **A. Vajpeyi**, G. Mentasti, Q. Baghi, O. Burke, and L. Speri, "An explicit and differentiable Wilson-Daubechies-Meyer transform for gravitational-wave data analysis", *arXiv e-prints*, Jun. 2026. arXiv: 2606.20269 `[gr-qc]`.

[4] **A. Vajpeyi**, R. Meyer, P. Maturana-Russell, and J. Liu, "Multivariate Bayesian P-spline estimation of spectral density matrices, with application to LISA TDI noise", *arXiv e-prints*, Jul. 2026. arXiv: 2607.04833 `[stat.ME]`.

[5] E. M. Zahraoui, P. Maturana-Russel, **A. Vajpeyi**, W. van Straten, R. Meyer, and S. Gulyaev, "Enhancing evidence estimation through informed probability density approximation", *Phys. Rev. D*, Apr. 2026, Also known as "MorphZ". arXiv: 2512.10283 `[stat.ME]`.

[6] T. Islam, **A. Vajpeyi**, F. H. Shaik, C.-J. Haster, V. Varma, S. E. Field, J. Lange, R. O'Shaughnessy, and R. Smith, "Analysis of GWTC-3 with fully precessing numerical relativity surrogate models", *Phys. Rev. D*, vol. 112, no. 4, Aug. 2025.

[7] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-4.0: An introduction to version 4.0 of the gravitational-wave transient catalog", *Astrophys. J. Lett.*, vol. 995, no. 1, p. L18, 2025, Art. no. L18.

[8] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run", *Astrophys. J.*, vol. 983, no. 2, p. 99, 2025, Art. no. 99.

[9] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for Gravitational Waves Emitted from SN 2023ixf", *Astrophys. J.*, vol. 985, no. 2, p. 183, 2025, Art. no. 183.

[10] J. Liu, **A. Vajpeyi**, R. Meyer, K. Janssens, J. E. Lee, P. Maturana-Russel, N. Christensen, and Y. Liu, "Variational inference for correlated gravitational wave detector network noise", *Phys. Rev. D*, vol. 111, no. 6, p. 062 003, Mar. 2025, Art. no. 062003.

[11] Team COMPAS, I. Mandel, J. Riley, A. Boesky, A. Brcek, R. Hirai, V. Kapil, M. Y. M. Lau, J. D. Merritt, N. Rodriguez-Segovia, I. Romero-Shaw, Y. Song, S. Stevenson, **A. Vajpeyi**, L. A. C. van Son, A. Vigna-Gomez, and R. Willcox, "Rapid stellar and binary population synthesis with COMPAS: Methods paper II", *Astrophys. J. Suppl.*, vol. 280, no. 1, p. 43, 2025, Art. no. 43. arXiv: 2506.02316 `[astro-ph.SR]`.

[12] LIGO Scientific Collaboration and Virgo Collaboration, "GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo during the First Half of the Third Observing Run", *Phys. Rev. D*, vol. 109, no. 2, p. 022 001, 2024, Art. no. 022001.

[13] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Observation of Gravitational Waves from the Coalescence of a 2.5-4.5 $M_\odot$ Compact Object and a Neutron Star", *Astrophys. J. Lett.*, vol. 970, no. 2, p. L34, 2024, Art. no. L34.

[14] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for gravitational-lensing signatures in the full third observing run of the LIGO–Virgo network", *Astrophys. J.*, vol. 970, no. 2, p. 191, 2024, Art. no. 191.

[15] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Constraints on the Cosmic Expansion History from GWTC-3", *Astrophys. J.*, vol. 949, no. 2, p. 76, 2023, Art. no. 76.

[16] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run", *Phys. Rev. X*, vol. 13, p. 041 039, 2023, Art. no. 041039.

[17] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Population of merging compact binaries inferred using gravitational waves through GWTC-3", *Phys. Rev. X*, vol. 13, no. 1, p. 011 048, 2023, Art. no. 011048.

[18] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for subsolar-mass black hole binaries in the second part of Advanced LIGO's and Advanced Virgo's third observing run", *Mon. Not. R. Astron. Soc.*, vol. 524, no. 4, pp. 5984–5992, 2023.

[19] LIGO Scientific Collaboration, Virgo Collaboration, KAGRA Collaboration, and GEO Collaboration, "Open data from the third observing run of LIGO, Virgo, KAGRA, and GEO", *Astrophys. J. Suppl.*, vol. 267, no. 2, p. 29, 2023, Art. no. 29.

[20] **A. Vajpeyi**, R. Smith, and E. Thrane, "Deep Follow-up for Gravitational-wave Inference: A Case Study with GW151226", *Astrophys. J.*, vol. 947, no. 1, p. 10, Apr. 2023, Art. no. 10. arXiv: 2203.13406 `[gr-qc]`.

[21] D. Williams, J. Veitch, M. L. Chiofalo, P. Schmidt, R. P. Udall, **A. Vajpeyi**, and C. Hoy, "Asimov: A framework for coordinating parameter estimation workflows", *Journal of Open Source Software*, vol. 8, no. 84, p. 4170, Apr. 2023.

[22] S. Y. Lehman, L. E. Christman, D. T. Jacobs, N. S. D. E. F. Johnson, P. Palchoudhuri, C. E. Tieman, **A. Vajpeyi**, E. R. Wainwright, J. E. Walker, I. S. Wilson, M. LeBlanc, L. W. McFaul, J. T. Uhl, and K. A. Dahmen, "Universal aspects of cohesion", *Granular Matter*, vol. 24, no. 1, p. 35, 2022, Art. no. 35.

[23] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "All-sky search for continuous gravitational waves from isolated neutron stars using Advanced LIGO and Advanced Virgo O3 data", *Phys. Rev. D*, vol. 106, no. 10, p. 102 008, 2022, Art. no. 102008.

[24] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "All-sky, all-frequency directional search for persistent gravitational waves from Advanced LIGO's and Advanced Virgo's first three observing runs", *Phys. Rev. D*, vol. 105, no. 12, p. 122 001, 2022, Art. no. 122001.

[25] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Constraints on dark photon dark matter using data from LIGO's and Virgo's third observing run", *Phys. Rev. D*, vol. 105, no. 6, p. 063 030, 2022, Art. no. 063030.

[26] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Narrowband searches for continuous and long-duration transient gravitational waves from known pulsars in the LIGO–Virgo third observing run", *Astrophys. J.*, vol. 932, no. 2, p. 133, 2022, Art. no. 133.

[27] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for continuous gravitational waves from 20 accreting millisecond X-ray pulsars in O3 LIGO data", *Phys. Rev. D*, vol. 105, no. 2, p. 022 002, 2022, Art. no. 022002.

[28] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for continuous gravitational-wave emission from the Milky Way center in O3 LIGO–Virgo data", *Phys. Rev. D*, vol. 106, no. 4, p. 042 003, 2022, Art. no. 042003.

[29] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for gravitational waves associated with gamma-ray bursts detected by Fermi and Swift during the LIGO–Virgo run O3b", *Astrophys. J.*, vol. 928, no. 2, p. 186, 2022, Art. no. 186.

[30] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for gravitational waves from Scorpius X-1 with a hidden Markov model in O3 LIGO data", *Phys. Rev. D*, vol. 106, no. 6, p. 062 002, 2022, Art. no. 062002.

[31] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Searches for gravitational waves from known pulsars at two harmonics in the second and third LIGO–Virgo observing runs", *Astrophys. J.*, vol. 935, no. 1, p. 1, 2022, Art. no. 1.

[32] B. McKernan, K. E. S. Ford, T. Callister, W. M. Farr, R. O'Shaughnessy, R. Smith, E. Thrane, and **A. Vajpeyi**, "LIGO–Virgo correlations between mass ratio and effective inspiral spin: testing the active galactic nuclei channel", *Mon. Not. R. Astron. Soc.*, vol. 514, no. 3, pp. 3886–3893, Jun. 2022. arXiv: 2107.07551 [astro-ph.HE].

[33] **A. Vajpeyi**, R. Smith, E. Thrane, G. Ashton, T. Alford, S. Garza, M. Isi, J. Kanner, T. J. Massinger, and L. Xiao, "A follow-up on intermediate-mass black hole candidates in the second LIGO–Virgo observing run with the Bayes Coherence Ratio", *Mon. Not. R. Astron. Soc.*, vol. 516, no. 4, pp. 5309–5317, Nov. 2022. arXiv: 2107.12109 [gr-qc].

[34] **A. Vajpeyi**, E. Thrane, R. Smith, B. McKernan, and K. E. S. Ford, "Measuring the Properties of Active Galactic Nuclei Disks with Gravitational Waves", *Astrophys. J.*, vol. 931, no. 2, p. 82, May 2022, Art. no. 82. arXiv: 2111.03992 [gr-qc].

[35] J. C. Bustillo, N. Sanchis-Gual, A. Torres-Forné, J. A. Font, **A. Vajpeyi**, R. Smith, C. Herdeiro, E. Radu, and S. H. W. Leong, "GW190521 as a Merger of Proca Stars: A Potential New Vector Boson of 8.7 $\times 10^{-13}$ eV", *Phys. Rev. Lett.*, vol. 126, no. 8, p. 081 101, Feb. 2021, Art. no. 081101. arXiv: 2009.05376 [gr-qc].

[36] LIGO Scientific Collaboration and Virgo Collaboration, "GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo during the First Half of the Third Observing Run", *Phys. Rev. X*, vol. 11, no. 2, p. 021 053, 2021, Art. no. 021053.

[37] LIGO Scientific Collaboration and Virgo Collaboration, "Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog", *Astrophys. J. Lett.*, vol. 913, no. 1, p. L7, 2021, Art. no. L7.

[38] LIGO Scientific Collaboration and Virgo Collaboration, "Search for lensing signatures in the gravitational-wave observations from the first half of LIGO–Virgo's third observing run", *Astrophys. J.*, vol. 923, no. 1, p. 14, 2021, Art. no. 14.

[39] LIGO Scientific Collaboration and Virgo Collaboration, "Tests of general relativity with binary black holes from the second LIGO–Virgo gravitational-wave transient catalog", *Phys. Rev. D*, vol. 103, no. 12, p. 122 002, 2021, Art. no. 122002.

[40] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "All-sky search for long-duration gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run", *Phys. Rev. D*, vol. 104, no. 10, p. 102 001, 2021, Art. no. 102001.

[41] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Constraints from LIGO O3 data on gravitational-wave emission due to r-modes in the glitching pulsar PSR J0537–6910", *Astrophys. J.*, vol. 922, no. 1, p. 71, 2021, Art. no. 71.

[42] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Observation of Gravitational Waves from Two Neutron Star-Black Hole Coalescences", *Astrophys. J. Lett.*, vol. 915, no. 1, p. L5, 2021, Art. no. L5.

[43] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Search for anisotropic gravitational-wave backgrounds using data from Advanced LIGO and Advanced Virgo's first three observing runs", *Phys. Rev. D*, vol. 104, no. 2, p. 022 005, 2021, Art. no. 022005.

[44] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Searches for continuous gravitational waves from young supernova remnants in the early third observing run of Advanced LIGO and Virgo", *Astrophys. J.*, vol. 921, no. 1, p. 80, 2021, Art. no. 80.

[45] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo's third observing run", *Phys. Rev. D*, vol. 104, no. 2, p. 022 004, 2021, Art. no. 022004.

[46] D. Williams, D. Macleod, **A. Vajpeyi**, and J. Clark, *transientlunatic/asimov: v0.3.2*, version v0.3.2, Jul. 2021.

[47] I. M. Romero-Shaw, et al., and **A. Vajpeyi**, "Bayesian inference for compact binary coalescences with BILBY: validation and application to the first LIGO-Virgo gravitational-wave transient catalogue", *Mon. Not. R. Astron. Soc.*, vol. 499, no. 3, pp. 3295–3319, Dec. 2020. arXiv: 2006.00714 `[astro-ph.IM]`.

[48] R. J. E. Smith, G. Ashton, **A. Vajpeyi**, and C. Talbot, "Massively parallel Bayesian inference for transient gravitational-wave astronomy", *Mon. Not. R. Astron. Soc.*, vol. 498, no. 3, pp. 4492–4502, Nov. 2020. arXiv: 1909.11873 `[gr-qc]`.

[49] M. Isi, R. Smith, S. Vitale, T. J. Massinger, J. Kanner, and **A. Vajpeyi**, "Enhancing confidence in the detection of gravitational waves from compact binaries using signal coherence", *Phys. Rev. D*, vol. 98, no. 4, p. 042 007, Aug. 2018, Art. no. 042007. arXiv: 1803.09783 `[gr-qc]`.

Talks

- **International Training Workshop on Frontiers from Quanta to Cosmos Physics, Beijing** — Aug–Sep '26
  Invited lecture for Master's and PhD students: Bayesian parameter estimation for compact binary coalescences in LIGO and LISA (accepted; visa pending).
- **AUT Mathematical Modelling and Analytics Symposium** — Nov '25
  Multivariate time-series spectral analysis with splines; applications to EEG, financial, and astrophysical data.
- **BayesComp25 Invited Session** — 2025
  Bayesian computation in astrophysics
- **COMPAS** — Nov '24
  Active Machine Learning with Deep GPs for COMPAS Population Inference
- **Flatiron Institute** — Jul '24
  Mind the Gaps — Time-Frequency Approaches for analysing LISA data with gaps
- **NZ Gravity Workshop, Canterbury** — Aug '23
  Tutorial on parallelisation techniques
- **NZ Summer School on Gravitational Waves** — Feb '23
  Bayesian parameter estimation for compact-binary inspiral signals
- **LIGO Rates and Populations** — Oct '21
  Measuring properties of AGN disks from BBH GW
- **FIT3170 Guest Lecture** — April '21
  UI/UX Design for programmers
- **OzGrav Online** — June '20
  Understanding our universe with parallel computing
- **OzGrav Retreat** — Nov '19
  A High Mass BH Search
- **Mt B. Observatory** — Feb '19
  GPU Computing in Astro
- **Google I/O Extended Bronx** — July '18
  "Icebreakers" tech demo

Outreach

- Assist UoA, Monash, OzGrav with public demos (open days, outreach, talks).
- Coded games Moonshot, three-body, looper, polarity, for outreach.

Public Talks

- Auckland Astronomical Society — Aug '25
  "Celestial Songs: A Cosmic Spacetime Symphony" — guest lecture, Stardome Observatory; recording.
- MOTAT STEM Fair — public engagement event — 2025
- Raising the Bar (Auckland) — Aug '24
  "Celestial Songs: A Cosmic Spacetime Symphony" — Auckland Fish Market; traced cosmic ripples from Einstein's 1916 theories to modern gravitational-wave discoveries and New Zealand's role. Event / podcast.
- Future Me (Auckland) — Jun '24
  "Thump in the night — Statistics and the hunt for black holes".
- Pint of Science (Melbourne) — 2019–22
  "Black Holes: The Ultimate Snitches" — how gravitational waves from colliding black holes let us study the dust-veiled centres of galaxies; delivered while a PhD candidate at Monash University.

Media

- 95bFM — Ready Steady Learn, radio interview — Aug '24
- RNZ — The science of space sounds, radio interview — Aug '24

Research Funding and Proposals

- **Dumont d'Urville NZ–France Science & Technology Support Programme** — 2026
  Principal Investigator; application pending.
- **ADACS Merit Allocation Program** (2 successful awards, ≈AUD $20K in-kind) — 2021, 2023
  One semester of professional software-development support per project: parallel_bilby and tess-atlas.

Hackathons

- Completed projects in 20 hackathons; published 19 games totaling +60Hrs of play.
- itch.io 1-BIT JAM #4 — Oct '24
  Towerfall: 1st in Gameplay, 4th of 331 overall
- Facebook Community Challenge ($2000) — Oct '20
  Best Oceania Region Award
- Google I/O Extended Bronx Hack: 1st Place ("Google Action") — July '18
- HackOH5: 1st Place ($1000) — April '17
  TensorFlow embeddings of newspaper text (1856-2010) demonstrating semantic change.

Teaching

Monash University: Teaching Assistant and Course Support (postgraduate and undergraduate)

- [FIT:3170] Software engineering practice — *team mentoring (independent); one guest lecture* 2021–22
  Independently mentored two agile teams of approximately 20 students each year. Monitored their use of the Scaled Agile Framework (SAFe), including documentation and meeting practices; advised on technical stacks; graded project work; recruited guest lecturers and external clients. Delivered a guest lecture on UI/UX design for programmers (Apr '21).
  2021 Monash teaching report: student satisfaction 88/100 raw and 98/100 weighted, both in the "Very High" band (SETU: exceeding expectations), against a Teaching Faculty average of 82/100. The survey went to the full 97-student cohort, of whom I taught roughly 40; 16 of the 17 respondents rated the teaching High or Very High. Report available on request.
- [FIT:9136] Algorithms and programming foundations in Python (postgraduate) — *tutorials (independent delivery); assessment co-authoring* 2020
  Sole tutor in the room for hybrid and online tutorials on Python and algorithms; co-authored assignments and examinations with the teaching team for the course lecturer's review; built automated grading and graded assessments. Tutorial content rotated weekly among three teaching assistants.
- [FIT:1048] Fundamentals of C++ — *laboratory tutorials (independent delivery)* July-Nov '19
  Ran laboratory sessions in C++ programming and walked students through code solutions.

Workshops and Short Courses — *designed and delivered independently*

- International Training Workshop on Frontiers from Quanta to Cosmos Physics, Beijing — *invited lecture (postgraduate)* Aug–Sep '26
  Bayesian parameter estimation for compact binary coalescences in LIGO and LISA; invitation accepted, visa pending.
- 2023 NZ Summer School on Gravitational Waves — *lecture and tutorial* Feb '23
  Sole designer and presenter of instruction on Bayesian parameter estimation for inspiral signals, to advanced undergraduate and postgraduate students.
- NZ Gravity Workshop, University of Canterbury — *tutorial* Aug '23
  Sole designer and presenter of a one-hour tutorial on parallelisation techniques, including multiprocessing and GPU programming.

The College of Wooster: Undergraduate Teaching Assistant — *laboratory and discussion sessions, delivered independently under a course instructor*

Awards

- RTP Stipend (AUD $28K) 2019-22
- Monash Tuition Scholarship (AUD $49K) 2019-22
- Wooster Commencement Speaker June '18
- The Jonas O. Notestein Prize — highest GPA in the graduating class
- The Arthur H. Compton Prize in Physics — highest standing in physics
- The Endowed Faculty Scholarship
- NCAA Honor Roll Dec '17
- MCURCSM: Best Paper Presentation Oct '17
- Dean's List and Scholarship '14-18
- Honour Societies: ΦΒΚ, ΠΜΕ June '17

References

Daniel Foreman–Mackey, Ph.D.
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Research Fellow, Monash University
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Student Supervision

Postgraduate — day-to-day research supervision alongside the students' official supervisors; not yet a formal co-supervisor appointment.

- Jianan Liu — PhD 2023–present
  University of Auckland; fast Bayesian methods (variational inference) for GW detector noise.
- Nazeela Aimen — PhD 2023–present
  University of Auckland; non-parametric Bayesian power spectral density estimation for LISA noise.
- Yukun Yang — MSc 2024
  University of Auckland; "Learning from Black Holes with Neural-net Surrogates".
- Jiaxuan (Steven) Zhou — MSc 2023
  Monash University; fast eccentric waveforms via reduced-order models and neural networks.
- Eyleen Amanda Sánchez García — MSc 2025
  Pontificia Universidad Católica de Valparaíso; CCSNe detection with normalising flows.
- Manfred Linton — Science Scholars 2023
  University of Auckland.

Undergraduate Capstone Team Mentoring (Monash FIT3170)

- OwlEyes 2021
  Final-year project team mentoring; Visual Testing for Android Apps.
- Automated grading tool 2021
  Final-year project team mentoring; FIT3170-project4-2021.
- MLVet 2022
  Final-year project team mentoring; video-editing app; MLVETDevelopers.
- RABIT 2022
  Final-year project team mentoring; FIT3170-FY-Project-7.